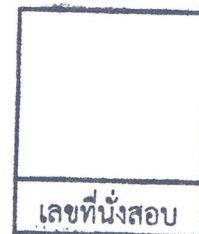


College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 2

Year: 2012

Subject: 341152 Electric Circuits II

Section: 5-6

Date: 7 January 2013

Time: 15.00-17.00

Name: _____ ID: _____ Class: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
 2. No documents are allowed to be taken out of the examination room.
 3. Text books are NOT allowed.
 4. Dictionaries and a calculator are allowed.
 5. Electronic communication devices are NOT allowed in the examination room.
 6. The examination has 11 pages (including this page), 3 parts (55 questions) and a total score of 71 points.
 7. Write all your answers on this exam paper and answer sheet.
-

Part 1 (25 points) True/False

Instruction: Mark the correct answer for following questions in the provided answer sheet.

1. The time constant for an inductor circuit equals its inductance divided by its resistance.
2. An ideal capacitor looks like a short circuit to dc current.
3. The steady-state level of the inductor current can be found by substituting its open-circuit equivalent.
4. The voltage across an inductor lags the current through it by 90° .
5. Inductive reactance increases inversely to frequency.
6. For a purely resistive element, the voltage and the current through the element are out of phase.
7. In parallel network, total admittance is the sum of the individual admittance.
8. The heavier the current flow for a given voltage, the larger the value for admittance.
9. Total impedance of a system is the sum of the individual impedances.
10. Admittance is the reciprocal of impedance.
11. Capacitance is directly proportional to the area of the plates.
12. Ideally, a capacitor does not dissipate energy.
13. The sinusoidal waveform is the only alternating waveform whose shape is unaffected by the response characteristics of R , L , and C elements.
14. For a series ac circuit with reactive elements, the voltages across a coil or capacitor are always in the same direction on a phasor diagram.

15. A capacitive network has a leading power factor.
16. The fact that the total impedance has a positive phase angle is a clear indication that the network is capacitive in nature.
17. The current divider rule for dc circuits cannot be used for ac circuits.
18. When the voltage divider rule is used for ac networks, the voltage must be expressed without the phase.
19. Power factor of resistive circuit is equal to one.
20. The magnitude of polar form is in RMS value.
21. When charging capacitor, the voltage of capacitor decreases exponentially.
22. When closing switch of RL DC circuit, electric current can flow in right at the beginning.
23. At very high frequency, inductor acts as an short circuit.
24. Resistor value is independent from frequency.
25. In RC series circuit, electric current leads the voltage at power supply.

Part 2 (26 points) Multiple choices

Instruction: Mark the correct answer for following questions in the provided answer sheet.

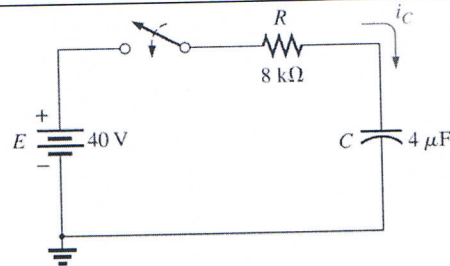


FIGURE 1

1. See Figure 1. What is the time constant in this circuit?
 (a) 12 s (b) 12 ms (c) 32 ks (d) 32 ms
2. See Figure 1. After closing the switch, what will the electric current be in the circuit at steady-state condition?
 (a) 0 A (b) 5 mA (c) 5 A (d) ∞ A

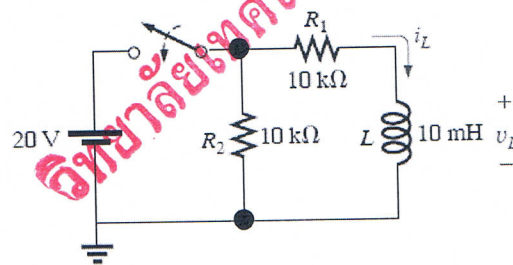


FIGURE 2

3. See Figure 2. After closing the switch, what will the voltage (v_L) across the inductor be after the circuit voltages and currents have reached steady-state values? Assume that the inductor is an ideal (lossless) device.
 (a) 0 V (b) 5 V (c) 10 V (d) 20 V

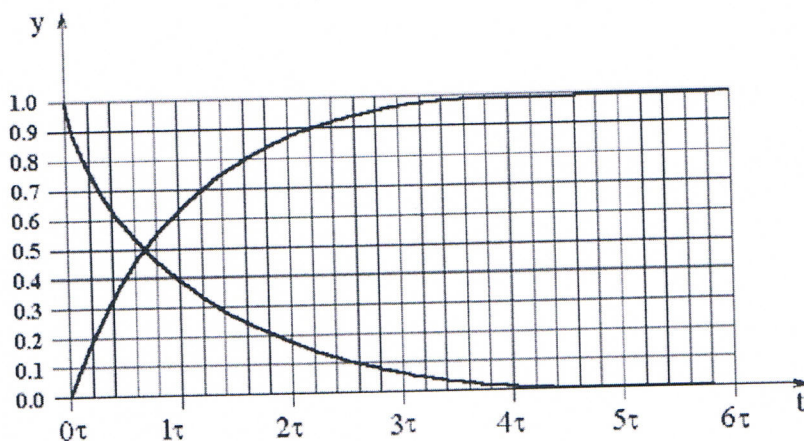


FIGURE 3

4. See Figure 3. These curves describe the voltage and current characteristics in an R-L circuit. Which of the following equations is decreasing curve?
- (a) $i_L(t) = e^{-t/\tau}$ (b) $v_L(t) = e^{-t/\tau}$
 (c) $i_L(t) = 1 - e^{-t/\tau}$ (d) $v_L(t) = 1 - e^{-t/\tau}$
5. See Figure 3. These curves describe the voltage and current characteristics in an R-L circuit. Which of the following equations is the increasing curve?
- (a) $i_L(t) = e^{-t/\tau}$ (b) $v_L(t) = e^{-t/\tau}$
 (c) $i_L(t) = 1 - e^{-t/\tau}$ (d) $v_L(t) = 1 - e^{-t/\tau}$
6. What is the inductive reactance at 0.5 Hz of a 1 H inductor?
- (a) $\pi \Omega$ (b) $2\pi \Omega$ (c) 1Ω (d) 2Ω
7. At what frequency does a $10/\pi \mu\text{F}$ capacitor have a reactance of 100Ω ?
- (a) 50 Hz (b) 500 Hz (c) 500 kHz (d) 5 MHz
8. If the voltage $v = 50\sin(500t)$ is impressed across a 25Ω resistor, which equation describes the resistor current?
- (a) $2\sin(500t + 90^\circ)$ (b) $2\cos(500t - 90^\circ)$
 (c) $2\sin(500t)$ (d) $2\sin(500t + 180^\circ)$
9. The voltage across a 100 mH coil is $v = 100 \sin 50t$. Which of these expressions describes the current?
- (a) $2000 \sin(50t - 90^\circ)$ (b) $20 \sin(50t + 90^\circ)$
 (c) $20 \sin 50t$ (d) $20 \sin(50t - 90^\circ)$

10. Which one of the following polar values is equivalent to $30 - j40$?
 (a) $50 \angle 36.9^\circ$ (b) $50 \angle 53.1^\circ$ (c) $50 \angle -36.9^\circ$ (d) $50 \angle -53.1^\circ$
11. Which one of the following values is equivalent to j^{10} ?
 (a) $1 \angle 0^\circ$ (b) $1 \angle 90^\circ$ (c) $1 \angle 180^\circ$ (d) $1 \angle -90^\circ$
12. Which one of the following values is equivalent to $-1 \angle 180^\circ$?
 (a) $1 \angle 0^\circ$ (b) $1 \angle 90^\circ$ (c) $1 \angle 180^\circ$ (d) $1 \angle -90^\circ$
13. Which one of the following rectangular values is equal to the polar form $20 \angle 55^\circ$?
 (a) $16.38 - j11.47$ (b) $16.38 + j11.47$
 (c) $11.47 - j16.38$ (d) $11.47 + j16.38$
14. Which one of the following phasor domain expressions is equivalent to the time domain expression $50\sin(\omega t + 15^\circ)$?
 (a) $70.7 \angle -15^\circ$ (b) $50 \angle 15^\circ$
 (c) $35.35 \angle 15^\circ$ (d) $70.7 \angle 15^\circ$

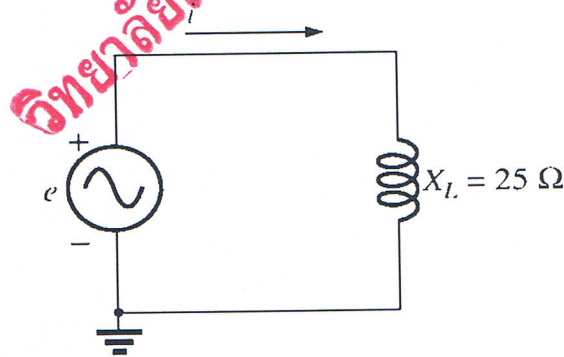


FIGURE 4

15. See Figure 4. What is the current i if e is $500 \sin(50t - 20^\circ)$?
 (a) $20 \sin(50t - 20^\circ)$ (b) $20 \sin(50t - 110^\circ)$
 (c) $20 \sin(50t + 110^\circ)$ (d) $20 \sin(50t + 70^\circ)$
16. A $10\mu F$ capacitor has a dc voltage of 200 V across its plates. What is the charge on the plates?
 (a) 20 mC (b) 2 mC (c) 3 mC (d) 13 mC

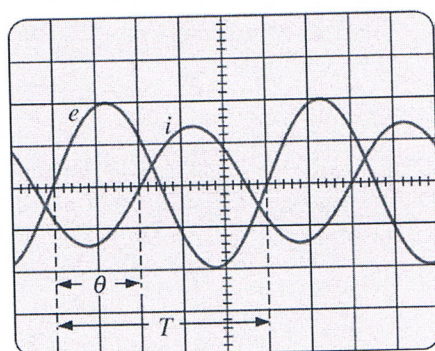
17. $v_C = 10(1 - e^{-t/30\mu s})$ Find v_C after 5 time constants.

(a) 7 V

(b) 9.93 V

(c) 8.89 V

(d) 10.1 V



Vertical sensitivity = 2 V/div.

Horizontal sensitivity = 0.2 ms/div.

FIGURE 5

18. See Figure 5. What relationship exists between voltages e and i ?

(a) e lags i by 36°

(b) e leads i by 36°

(c) e lags i by 144°

(d) e leads i by 114°

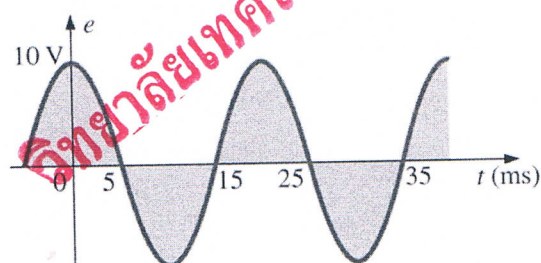


FIGURE 6

19. See Figure 6. What is the frequency of the waveform?

(a) 15 Hz

(b) 25 Hz

(c) 50 Hz.

(d) 100 Hz

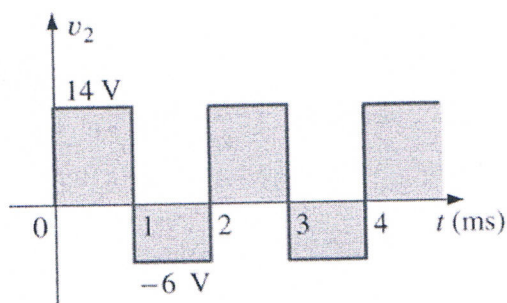


FIGURE 7

20. See Figure 7. What is the average value of the waveform?

- (a) 0 V (b) 2 V (c) 4 V (d) 8 V

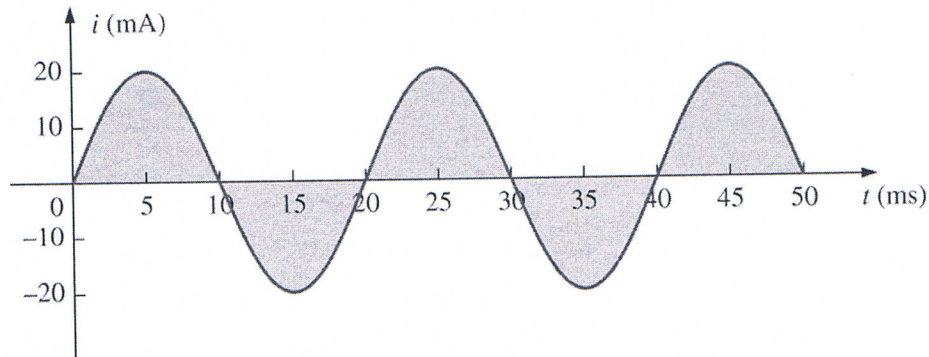


FIGURE 8

21. See Figure 8. What is the RMS value of the waveform?

- (a) 10 V (b) 12.72 V (c) 14.14 V (d) 28.28 V

22. See Figure 8. What is the average value of the waveform?

- (a) 0 V (b) 0.636 V (c) 6.36 V (d) 12.72 V

23. Find the positive peak value of the waveform in Figure 9.

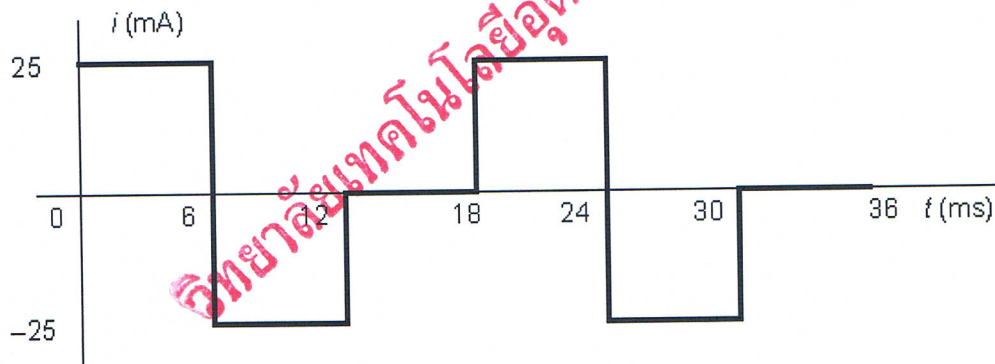


FIGURE 9

- (a) 50 mA (b) 12.5 mA (c) 25 mA (d) 20 mA

24. Using the waveform in Figure 9, find the frequency of the current.

- (a) 83.3 Hz (b) 167 Hz (c) 55.6 Hz (d) 0.083 Hz

25. Express the phasor $100 \angle 60^\circ$ as a sine wave with a frequency of 60 Hz.

- (a) $141.4 \sin(60t + 60^\circ)$ (b) $100 \sin(377t + 60^\circ)$
(c) $141.4 \sin(377t + 60^\circ)$ (d) $100 \sin(60t + 60^\circ)$

26. Evaluate the expression $(4 + j5) \times (39 \angle -42^\circ)$ and express the answer in polar form.

- (a) $79.5 \angle 9.3^\circ$ (b) $79.5 \angle -9.3^\circ$ (c) $249.6 \angle 9.3^\circ$ (d) $249.6 \angle -9.3^\circ$

Part 3 (20 points) Calculation

Instruction: Show the mathematical expression and answers the following problems.

1. Find the total ADMITTANCE (Y_T) and IMPEDANCE (Z_T) of the circuit of Figure 10.

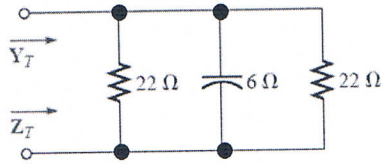


FIGURE 10

2. See Figure 11. Determine the energy stored by capacitor (W_C) and inductor (W_L).

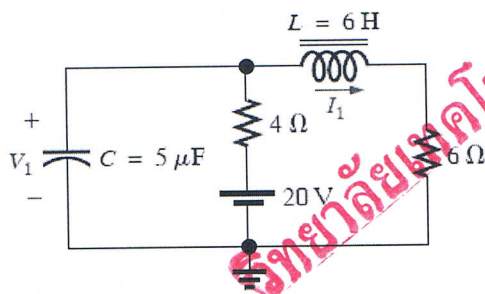


FIGURE 11

3. Find the average power loss (P) and power factor $\{\cos(\theta)\}$ for a circuit whose input current and voltage are as $v(t) = 10 \sin(\omega t)$ [V] and $i(t) = 2 \sin(\omega t - 45^\circ)$ [A].

4. Find V_1 and V_2 from Figure 12.

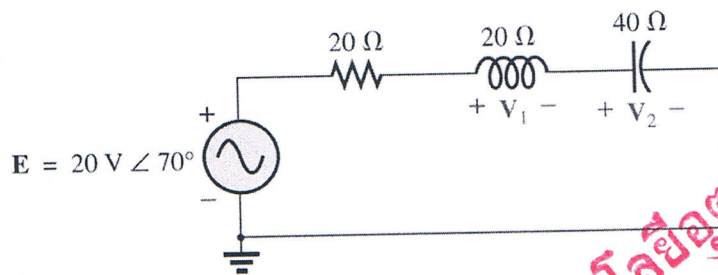


FIGURE 12

Instructors: Asst. Prof. Dr. Poolsak Koseeyaporn

Asst. Prof. Dr. Titipong Lertwiriaprapa

Answer Sheet

Name: _____ ID: _____

Subject: 341152 Electric Circuits II

เลขที่นั่งสอบ

Part 1

	True	False
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Part 2

	a	b	c	d
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